Timeline Granularity Transformation: Aligning Narrative Logic with Temporal-Semantic Evolution

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Abstract. Dynamic-granularity timeline summarization has shown that the same news topic may admit multiple valid levels of detail. We study a distinct problem, **Timeline Granularity Transformation (TGT)**. Given an already available fine-grained timeline, the goal is to produce a single user-requested K-node timeline by selecting K source nodes in chronological order while preserving the temporal-semantic evolution of the full event trajectory. Unlike multi-granularity generation from raw documents, TGT is a post-hoc transformation problem over an existing event sequence; unlike constrained timeline summarization, the control signal specifies global granularity rather than a topical or aspectual slice. The main challenge is structure-preserving compression under non-homogeneous event dynamics. Because real news streams alternate between burst phases and dormant phases, methods based on local salience or near-uniform spacing often distort event evolution. We propose **Time-GRPO**, a sequence-level reinforcement learning framework with three rewards. STAR promotes temporal-semantic pacing. A density-aware temporal reward adapts to non-uniform event distributions. An entropy-regularized information reward balances coverage and redundancy under tight budgets. Experiments on DTELS-Bench and CCKS2025 at the benchmark-supported settings $K = 5$ and $K = 10$ show that Time-GRPO achieves the highest STAR across datasets and backbones. Comparing against a non-RL control with the same reward further shows that these gains come from sequence-level policy optimization rather than reward design alone. The gains are largest on long-horizon timelines, while burst-heavy streams remain a boundary case in which explicit clustering is competitive on exact-match node selection.

1 Introduction. Timeline Summarization (TLS) helps readers digest long-running news events by organizing them into chronological event skeletons [22, 13, 27, 4, 11]. As events unfold over months or years, however, complete timelines can become too long to read efficiently. The desired level of detail is also user-dependent. A casual reader may need a 5-node overview, while a domain expert may prefer a 10-node timeline that preserves finer causal structure [32]. This motivates **Timeline Granularity Transformation (TGT)**, where the goal is to compress a fine-grained timeline into a shorter one under a user-specified budget while preserving the temporal-semantic evolution of the original event trajectory.

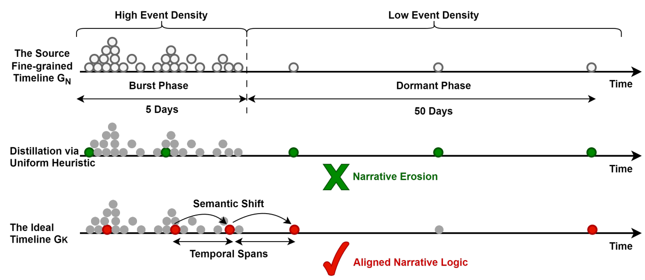

Figure 1.1: Timeline Granularity Transformation (TGT) under non-homogeneous event dynamics. **Top:** The source timeline $\mathcal{G}_N$ contains a 5-day burst phase (high event density) and a 50-day dormant phase (low event density). **Middle:** A uniform-spacing heuristic treats all intervals equally and causes narrative erosion. **Bottom:** An ideal TGT aligns semantic shifts with temporal spans within a K-node budget.

TGT is related to dynamic-granularity timeline summarization (DTELS) [32] and Constrained Timeline Summarization (CTLS) [18], but it is not the same as either of them. DTELS studies document-to-timeline generation: given a topic, a document set, and a granularity indicator, the model generates a timeline at the requested level of detail. TGT instead starts from an already available fine-grained timeline and transforms it into one target K-node timeline at inference time. The output is not free-form generation. Each selected node must come directly from the source timeline, remain in chronological order, and satisfy an exact budget. CTLS differs in another respect. There, the constraint specifies which aspect or perspective of a topic should be summarized from the document set. In TGT, the control signal specifies the global granularity of the full timeline. The task is therefore to compress the whole event trajectory into a coarser but faithful representation, rather than summarize a thematic slice.

A central challenge in TGT is *narrative erosion*. The output stays chronological, yet it distorts how the event evolves over time. Real news streams are highly non-homogeneous, alternating between bursty periods and long dormant periods [30]. Many existing methods, by contrast, either score events independently or spread selections too evenly over time [2, 9]. As Figure 1.1 shows, this bias thins out critical transitions during bursts and over-allocates budget to low-information intervals during dormancy.

We formulate TGT as a budget-conditioned transformation problem in which the target timeline is formed by selecting a chronological subsequence from the source timeline. Given a source timeline $\mathcal{G}_N$, the goal is to choose a K-node subsequence $\mathcal{G}_K$ that preserves temporal-semantic evolution. The key requirement is **temporal-semantic alignment**. Large semantic shifts should be reflected by suitably large temporal spans, while dense phases should be represented without wasting budget on redundant updates. This makes TGT a sequence-level compression problem rather than ranking individual nodes by local importance.

Existing methods remain fragile under tight budgets such as $K = 5$ and $K = 10$ [18, 28]. We see two main reasons. First, many extractive heuristics and zero/few-shot LLMs have a uniform-spacing bias at inference time. They assess events in isolation or distribute selections too evenly over the timeline, ignoring the underlying event density [26]. Second, standard supervised fine-tuning is poorly matched to TGT. Reference timelines tell the model which nodes were selected, but not why certain temporal gaps, burst phases, or redundancies matter. As a result, SFT tends to imitate surface patterns and may spend scarce budget on semantically repetitive updates [24, 17].

To address these issues, we propose **Time-GRPO**, a reinforcement learning framework that optimizes sequence-level timeline quality rather than token-level imitation. Time-GRPO uses three rewards. **STAR** encourages temporal-semantic pacing. A **density-aware temporal reward** adapts selection to the source event distribution. An **entropy-regularized information reward** balances coverage and redundancy under extreme compression. Because the action space is discrete and combinatorial, with size $\binom{N}{K}$, we further introduce an **adaptive curriculum** that guides training from structural validity to content anchoring and then to temporal-semantic refinement.

Our contributions are summarized as follows:

- We formulate **Timeline Granularity Transformation (TGT)** as a budget-conditioned timeline-to-timeline transformation problem with exact-cardinality constraints, where the output is a chronological subsequence of the source timeline, and identify temporal-semantic alignment as the key requirement for preserving event evolution under compression.

- We propose **Time-GRPO**, a sequence-level reinforcement learning framework with three rewards—STAR, a density-aware temporal reward, and an entropy-regularized information reward—to transform fine-grained timelines under non-homogeneous event dynamics.

- We introduce an **adaptive curriculum** to stabilize policy optimization in the combinatorial action space. On DTELS-Bench and CCKS2025 at the benchmark-supported budgets $K = 5$ and $K = 10$, Time-GRPO achieves the strongest STAR across datasets and backbones; comparisons with a same-reward non-RL control show that the gains come from sequence-level policy optimization rather than reward design alone.

2 Problem Formulation.

We formalize **Timeline Granularity Transformation (TGT)** as a budget-conditioned timeline-to-timeline transformation problem with exact-cardinality and strict-subsequence constraints. Given an already available fine-grained source timeline, the goal is to transform it into a shorter K-node target timeline while preserving the temporal-semantic evolution of the original event trajectory.

2.1 Task Formulation

Let the fine-grained source timeline be a time-ordered sequence $\mathcal{G}_N = (e_1, e_2, \ldots, e_N)$, where each event node is defined as $e_i = \langle t_i, c_i \rangle$, with timestamp t_i and textual summary c_i. The target timeline is a K-node sequence $\mathcal{G}_K = (e_{p_1}, e_{p_2}, \ldots, e_{p_K})$, such that $1 \le p_1 < p_2 < \cdots < p_K \le N$.

Because each selected node must be copied from the source timeline and remain in chronological order, $\mathcal{G}_K$ is a strict subsequence of $\mathcal{G}_N$. The control signal in this paper is an explicit user-specified budget K. Although granularity could in principle be expressed numerically or textually, all experiments here instantiate it as $K \in \{5, 10\}$, which are the benchmark-supported settings in DTELS-Bench and CCKS2025. Smaller budgets produce coarser timelines, whereas larger budgets retain finer event detail.

A good target timeline should preserve temporal-semantic evolution. We operationalize this requirement through temporal-semantic alignment between adjacent selected events. Let $E(\cdot)$ denote a pre-trained semantic embedding function that maps a textual summary c to a dense vector in $\mathbb{R}^d$. We define the semantic shift between adjacent selected nodes as

$$\Delta S_j = 1 - \cos\big(E(c_{p_j}), E(c_{p_{j+1}})\big), \tag{2.1}$$

and the corresponding temporal span as

$$\Delta T_j = t_{p_{j+1}} - t_{p_j}. \tag{2.2}$$

A target timeline exhibits temporal-semantic alignment when the pattern of semantic shifts is well aligned with the pattern of temporal spans across adjacent selected nodes.

2.2 Optimization Objective

We cast TGT as a policy optimization problem. Let $\pi_\theta(\mathcal{G}_K \mid \mathcal{G}_N, K)$ denote a stochastic policy parameterized by θ, which maps a source timeline and a target budget to a distribution over valid K-node target timelines. The policy is trained to maximize a composite reward for sequence-level timeline quality.

For notational simplicity, we write this reward as $R(\mathcal{G}_K \mid \mathcal{G}_N, K)$. Some reward components additionally depend on the gold reference $\mathcal{G}_{\text{Ref}}$ during training, as detailed later in Section 4.1.

The optimal policy θ^* is obtained by maximizing the expected reward:

$$\theta^* = \arg\max_\theta \mathbb{E}_{\mathcal{G}_K \sim \pi_\theta(\cdot|\mathcal{G}_N, K)} \left[R(\mathcal{G}_K \mid \mathcal{G}_N, K) \right]. \tag{2.3}$$

3 Methodology. To address the inherent challenges of Timeline Granularity Transformation, we propose **Time-GRPO**, a reinforcement learning framework designed to align temporal spans with semantic narratives.

3.1 Time-GRPO Framework. **Time-GRPO** fundamentally shifts the paradigm from supervised fitting to outcome-based exploration. As illustrated in Figure 3.1, we formulate the timeline transformation as a policy optimization problem, employing a Large Language Model (LLM) as the policy network π_θ to navigate the high-dimensional combinatorial search space $\binom{N}{K}$.

To efficiently optimize π_θ, we utilize Group Relative Policy Optimization (GRPO) as the training backbone. Unlike traditional actor-critic methods like PPO [19], which necessitate a separate value network for advantage estimation, GRPO **eliminates the memory overhead of a parameterized critic**. Instead, it adopts a memory-efficient paradigm that estimates the baseline using the average reward of a group of sampled outputs. Concretely, for each input $\mathcal{G}_N$, the policy samples a group of candidate outputs $\{\mathcal{G}_K^{(1)}, \ldots, \mathcal{G}_K^{(G)}\}$ and updates the model based on relative advantages derived from the group mean. This process reinforces global narrative properties by maximizing a composite reward that harmonizes narrative logic, event density, and informativeness. The detailed mathematical formulations are provided in Appendix A.

3.2 Composite Reward Functions. To steer the policy network π_θ towards the optimal subsequence $\mathcal{G}_K$, we design a multi-objective reward mechanism that serves as a proxy for the optimization goals defined in Section 2. The total reward is formulated as a weighted aggregation of three task rewards plus one format-constraint reward, harmonizing temporal-semantic alignment, density-aware pacing, informativeness, and format validity:

$$R_{\text{total}} = \omega_1 R_{\text{STAR}} + \omega_2 R_{\text{Time}} + \omega_3 R_{\text{Info}} + \omega_4 R_{\text{Format}}, \tag{3.1}$$

where ω_i denote scaling coefficients that are dynamically modulated via curriculum learning (detailed in §3.3). For readability, the main text presents the reward design at a symbolic level; the exact coefficient settings and operational details used in experiments are deferred to Appendix B. Some reward components additionally depend on the gold reference $\mathcal{G}_{Ref}$ during training, whereas inference uses only the prompt and source timeline $\mathcal{G}_N$.

3.2.1 Semantic-Temporal Alignment Reward (STAR). Building on the principle of temporal-semantic evolution introduced in Section 2, narrative logic requires semantic shifts to align with temporal spans. Existing approaches often suffer from narrative erosion by treating all time intervals too uniformly. To counteract this, we propose the **Semantic-Temporal Alignment Reward (STAR)**, which operationalizes the premise that the distribution of semantic change should align with the distribution of elapsed time. To avoid allowing extremely long dormant gaps to dominate the signal, we apply a log transform to temporal intervals before normalization.

Formally, let $\Delta T_j = t_{p_{j+1}} - t_{p_j}$ and $\Delta S_j = 1 - \cos(E(c_{p_j}), E(c_{p_{j+1}}))$. We define the log-scaled normalized temporal span τ_j and normalized semantic divergence s_j as follows:

$$\begin{aligned} \tilde{\Delta T}_j &= \log(1 + \Delta T_j), \\ \tau_j &= \frac{\tilde{\Delta T}_j}{\sum_{m=1}^{K-1} \tilde{\Delta T}_m}, \\ s_j &= \frac{\Delta S_j}{\sum_{m=1}^{K-1} \Delta S_m}. \end{aligned} \tag{3.2}$$

Here, $E(\cdot)$ denotes a fixed pre-trained semantic embedding function, and cosine distance in this embedding space serves as a proxy for semantic change.

The STAR score used for both training-time reward computation and post-hoc evaluation is then formulated as

$$R_{\text{STAR}} = 1 - \frac{1}{2} \sum_{j=1}^{K-1} |\tau_j - s_j|. \tag{3.3}$$

The inner term is the total variation distance between the normalized temporal and semantic distributions, so $R_{\text{STAR}} \in [0, 1]$, with larger values indicating better alignment. By rewarding higher alignment between $\boldsymbol{\tau}$ and s, STAR discourages narrative stagnation (large temporal gaps with limited semantic change) and abrupt disjoints (large semantic jumps compressed into short spans), thereby promoting coherent narrative pacing. In implementation, if fewer than two nodes are selected or either normalization becomes degenerate, we set $R_{\text{STAR}} = 0$; for robustness, negative temporal gaps are clipped to zero before the log transform.

3.2.2 Density-Aware Temporal Reward. The occurrence of real-world events is inherently heterogeneous, frequently alternating between dormant phases and sudden bursts of activity. Motivated by point-process views of burstiness, we design the temporal reward R_{Time} as a composite metric that balances reference-based anchoring and density-aware adaptation:

$$R_{\text{Time}} = \alpha \cdot R_{\text{align}} + (1 - \alpha) \cdot R_{\text{adapt}}, \tag{3.4}$$

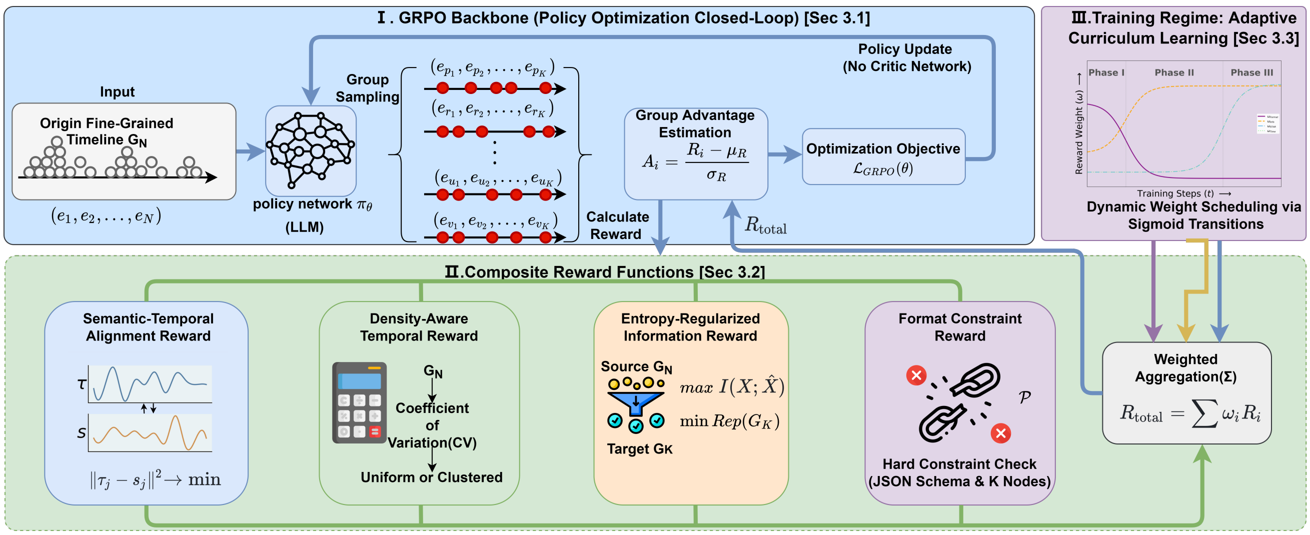

Figure 3.1: Overview of the Time-GRPO framework for Timeline Granularity Transformation (TGT).

where $\alpha \in [0, 1]$ is a balancing coefficient. Here, R_{align} is a training-time anchoring term that matches the temporal pacing of the generated timeline to the reference timeline, while R_{adapt} is a density-aware term defined on the generated and source timelines.

We categorize the source $\mathcal{G}_N$ into Uniform ($\mathcal{U}$) or Clustered ($\mathcal{C}$) states using a multi-indicator voting mechanism based on dispersion statistics (detailed in Appendix B.1). The adaptive reward is then formulated as:

(3.5)
$$R_{\text{adapt}} = \begin{cases} 1 - \min\left(1, \frac{\text{CV}(\mathcal{G}_K)}{\sigma_{\text{th}}}\right) & \text{if } \mathcal{G}_N \in \mathcal{U}, \\ \lambda \cdot \text{Span}(\mathcal{G}_K) + (1 - \lambda) \cdot \text{Sim}_{\text{ZR}} & \text{if } \mathcal{G}_N \in \mathcal{C}, \end{cases}$$

where $\text{CV}(\cdot)$ denotes the coefficient of variation, σ_{th} is a tolerance threshold for variance in uniform regimes, $\text{Span}(\mathcal{G}_K)$ denotes the normalized temporal coverage of the selected timeline relative to the source stream, and $\text{Sim}_{\text{ZR}} = 1 - |\text{ZR}(\mathcal{G}_K) - \text{ZR}(\mathcal{G}_N)|$ measures the similarity of zero-interval ratios. The exact operationalization of R_{align} and the regime-specific implementation details of R_{adapt} are provided in Appendix B.1.

3.2.3 Entropy-Regularized Information Reward. Given the extreme compression scenario where $N \gg K$, each selected node constitutes a critical allocation of the limited information budget. To ensure that the distilled timeline $\mathcal{G}_K$ is both representative and concise, we ground the design of R_{Info} in an entropy-regularized coverage objective:

(3.6)
$$\begin{aligned} R_{\text{Info}} = \mu_1 \cdot & \underbrace{(\text{ROUGE}(\mathcal{G}_K, \mathcal{G}_N) \odot \mathcal{M}_{\text{Rep}}(\mathcal{G}_K))}_{\text{Penalized Relevance}} \\ & + \mu_2 \cdot \underbrace{H_{\text{norm}}(\mathcal{G}_K; \mathcal{G}_{Ref})}_{\text{Diversity}}, \end{aligned}$$

where μ_1, μ_2 are balancing coefficients.

This objective has three roles. First, $\text{ROUGE}(\mathcal{G}_K, \mathcal{G}_N)$ serves as a proxy for source coverage. Second, the repetition gate $\mathcal{M}_{\text{Rep}}(\mathcal{G}_K)$ suppresses redundant selections under a strict budget. Third, the normalized entropy term $H_{\text{norm}}(\mathcal{G}_K; \mathcal{G}_{Ref})$ encourages lexical diversity while using the reference only for training-time normalization. The exact implementation of $\mathcal{M}_{\text{Rep}}$ and H_{norm} is given in Appendix B.2.

3.2.4 Format Constraint Reward. The combinatorial search space contains numerous invalid states that violate the structural requirements of the task. To enforce strict adherence to the output specifications, we implement a hard constraint mechanism via R_{Format}. This mechanism imposes a penalty $\mathcal{P}$ on any output that fails to satisfy the predefined JSON schema or violates the exact cardinality constraint K. The format reward is formally defined as:

(3.7)
$$R_{\text{Format}} = \begin{cases} 1 & \text{if } \mathcal{G}_K \text{ satisfies all constraints} \\ 1 - \mathcal{P} & \text{otherwise,} \end{cases}$$

where the penalty coefficient $\mathcal{P} \in [0, 1]$ ensures that the reward remains non-negative. This bounded reward structure prevents the model from suffering from gradient instability due to excessive negative feedback during the initial exploration phase, thereby facilitating smoother convergence toward the valid action subspace.

Total Reward Aggregation. The final optimization objective is computed as a dynamic weighted aggregation of the four components (R_{STAR}, R_{Time}, R_{Info}, and R_{Format}). These weights are adaptive rather than static. They are modulated through an **Adaptive Curriculum Learning** strategy, which progressively steers the model from basic structural alignment to advanced narrative refinement.

Table 4.1: Dataset statistics after cleaning. CR denotes the average compression ratio ($|\mathcal{G}_N|/K$).

Dataset	K	Topics	Avg. $\|\mathcal{G}_N\|$	Days	CR
DTELS-Bench	5	373	31.75	1049.97	6.35
	10	362			3.18
CCKS2025	5	289	34.02	23.37	6.80
	10	289			3.40

3.3 Training Regime: Adaptive Curriculum Learning. Directly optimizing the composite reward R_{total} can lead to convergence failure because hard structural constraints and high-level narrative goals become useful at different stages of learning. To stabilize the optimization of π_θ, we adopt an adaptive curriculum learning strategy based on sigmoid-modulated weight transitions:

$$\omega_c(t) = \omega_{\text{init}} + \frac{\omega_{\text{target}} - \omega_{\text{init}}}{1 + e^{-\gamma\cdot(t-\tau_c)}}, \tag{3.8}$$

where ω_{init} and ω_{target} denote the initial and target weights for reward component c, τ_c is the phase transition center, and γ controls the transition slope. The exact scheduling parameters used in experiments are provided in Appendix B.5.

This curriculum induces three logical phases:

Phase I: Format Alignment ($0 \le t < \tau_1$). Training begins with a dominant weight on R_{Format}, forcing the policy into the valid action subspace before higher-level reasoning is activated.

Phase II: Content Anchoring ($\tau_1 \le t < \tau_2$). Once structural validity is achieved, the weight on R_{Format} gradually decays while the weight on R_{Info} increases. This stage encourages high-coverage, low-redundancy selections before temporal-semantic refinement.

Phase III: Narrative Refinement ($t \ge \tau_2$). In the final stage, the weights on R_{STAR} and R_{Time} are strengthened to refine temporal-semantic alignment and density-aware pacing under non-homogeneous event streams.

4 Experiments.

4.1 Experimental Setup. This section details the datasets, baselines, and evaluation metrics used to evaluate our proposed framework.

Dataset. We evaluate our framework on two benchmarks: DTELS-Bench (long-term evolution) and CCKS2025 Task 6 (high-density event bursts)[1]. To ensure the rigor of the Timeline Granularity Transformation (TGT) task, we filtered the datasets by removing redundant entries and ensuring that each target timeline $\mathcal{G}_K$ is a chronological subsequence of the fine-grained source timeline $\mathcal{G}_N$. For a rigorous and fair comparison with existing baselines, our experiments specifically evaluate outputs at $K = 5$ and $K = 10$. These exact values were not chosen arbitrarily; rather, they strictly adhere to the established standard evaluation protocols of the DTELS-Bench and CCKS2025 datasets. The refined statistics are summarized in Table 4.1. Accordingly, we restrict quantitative evaluation to these benchmark-supported settings and do not report unseen-K generalization, as no benchmark-standard gold references are available beyond K=5 and K=10 for the current datasets.

Table 4.2: Reward dependency summary. Terms involving $\mathcal{G}_{Ref}$ are used only during training. During decoding, the model uses only the prompt and source timeline $\mathcal{G}_N$; STAR is additionally computed post hoc as an evaluation metric.

Reward Term	Dependency
R_{STAR}	generated $\mathcal{G}_K$ only
R_{Time} (R_{align})	$\mathcal{G}_K + \mathcal{G}_{Ref}$
R_{Time} (R_{adapt})	$\mathcal{G}_K + \mathcal{G}_N$
R_{Info} (ROUGE$\times M_{\text{Rep}}$)	$\mathcal{G}_K + \mathcal{G}_N$
R_{Info} (H_{norm})	$\mathcal{G}_K + \mathcal{G}_{Ref}$
R_{Format}	generated $\mathcal{G}_K$ only

Baselines. We compare Time-GRPO against **HC-Centroid**, a statistical baseline utilizing hierarchical clustering with spatiotemporal distance [23]. For LLMs, we evaluate **Llama3.1-8B** [6] and **Qwen3-8B** [31] across five settings: **Zero-shot** (with **GPT-5.2**[2] as a strong proprietary reference), **SFT** (standard supervised fine-tuning), **RS-SFT** (Reward-Search SFT, a same-reward non-RL control), **DPO** (preference optimization using ROUGE/STAR-based pairs), and **Time-GRPO**. In RS-SFT, we approximate the same composite objective used by Time-GRPO without policy optimization: on the training split, candidate K-node timelines are scored by the composite reward and the highest-scoring candidate is used as a pseudo target for teacher-forcing fine-tuning. At inference time, RS-SFT uses standard decoding and does not access reward signals or gold references. This design disentangles the contribution of reward design from the contribution of online sequence-level policy optimization. Hyperparameters are detailed in Appendix B.

Evaluation Metrics. We evaluate timeline quality using: (1) **Node Accuracy (Acc)**, measuring precision by requiring exact timestamp t_i and content c_i matches with the ground truth; (2) **Semantic Coverage (ROUGE)**, utilizing ROUGE-1/2 (F1) to assess the preservation of core information; and (3) **Temporal-Semantic Alignment (STAR)**, a sequence-level metric that measures the alignment between temporal spans and semantic shifts. Crucially, we position STAR as a complementary indicator of narrative pacing rather than a standalone measure of timeline quality, as it must be jointly evaluated with Acc and ROUGE to rule out reward hacking (detailed correlation analysis is provided in Appendix C).

[1] https://tianchi.aliyun.com/competition/entrance/532361

[2] https://openai.com/index/introducing-gpt-5.2

Table 4.3: Main results of Timeline Granularity Transformation on DTELS-Bench and CCKS2025. Acc denotes accuracy, and R-1/2 refer to ROUGE-1 and ROUGE-2 respectively. The best results per backbone are in **bold**. The best results across all settings are underlined.

Dataset	Backbone	Setting	K = 5				K = 10			
			Acc	R-1	R-2	STAR	Acc	R-1	R-2	STAR
DTELS-Bench (Long-term)	*Statistical*	HC-Centroid	**0.306**	**0.291**	**0.123**	**0.544**	**0.513**	**0.493**	**0.302**	**0.456**
	GPT-5.2	Zero-shot	**0.432**	**0.357**	**0.185**	**0.584**	**0.609**	**0.577**	**0.258**	**0.462**
	Llama3.1-8B	Zero-shot	0.378	0.309	0.130	0.501	0.534	0.582	0.347	0.367
		SFT	0.396	**0.401**	**0.190**	0.580	0.547	0.446	0.226	0.393
		RS-SFT	0.408	0.356	0.176	0.610	0.549	0.494	0.249	0.443
		DPO	0.422	0.345	0.166	0.554	0.555	0.508	0.266	0.398
		Time-GRPO	**0.446**	0.345	0.188	**0.621**	**0.634**	**0.623**	**0.406**	**0.514**
	Qwen3-8B	Zero-shot	0.384	0.301	0.151	0.500	0.534	0.446	0.226	0.333
		SFT	0.388	0.301	0.172	0.507	0.540	0.484	0.238	0.340
		RS-SFT	0.437	0.339	0.149	0.582	0.568	0.557	0.289	0.479
		DPO	0.407	0.341	0.175	0.572	0.576	0.496	0.252	0.441
		Time-GRPO	**0.447**	**0.373**	**0.186**	**0.631**	**0.616**	**0.583**	**0.289**	**0.542**
CCKS2025 (High-density)	*Statistical*	HC-Centroid	**0.260**	**0.364**	**0.145**	**0.590**	**0.446**	**0.522**	**0.289**	**0.479**
	GPT-5.2	Zero-shot	**0.153**	**0.306**	**0.115**	**0.606**	**0.299**	**0.423**	**0.216**	**0.417**
	Llama3.1-8B	Zero-shot	0.159	0.255	0.075	0.466	0.239	0.375	0.200	0.337
		SFT	0.158	0.219	0.077	0.450	0.250	0.299	0.144	0.392
		RS-SFT	0.163	**0.285**	**0.103**	0.371	0.267	**0.428**	0.199	0.340
		DPO	0.165	0.247	0.089	0.548	0.268	0.364	0.180	0.475
		Time-GRPO	**0.172**	0.266	0.097	**0.645**	**0.291**	0.427	**0.211**	**0.558**
	Qwen3-8B	Zero-shot	0.145	0.270	0.102	0.550	0.275	0.355	0.175	0.249
		SFT	0.144	0.252	0.093	0.556	0.268	0.379	0.191	0.297
		RS-SFT	0.147	**0.293**	**0.129**	0.448	0.261	0.414	0.197	0.381
		DPO	0.158	0.247	0.103	0.555	0.223	0.371	0.186	0.386
		Time-GRPO	**0.173**	0.260	0.126	**0.668**	**0.336**	**0.441**	**0.235**	**0.633**

Implementation Notes. We instantiate the semantic embedding function $E(\cdot)$ with **m3e-base** [29] and keep it frozen throughout training and evaluation. During decoding, the model conditions only on the prompt and source timeline $\mathcal{G}_N$; no reward or gold reference is exposed to the model, and $\mathcal{G}_{Ref}$ appears only in the training-time components summarized in Table 4.2.

4.2 Main Results. Table 4.3 reports results on two benchmarks with markedly different temporal regimes. As the dataset statistics in Table 4.1 show, DTELS-Bench spans much longer horizons than CCKS2025 (1049.97 vs. 23.37 days on average after cleaning). We therefore interpret the results not as a single leaderboard, but as evidence across two complementary settings. We discuss them from three perspectives.

Overall Performance. The clearest pattern in Table 4.3 is that Time-GRPO consistently improves sequence-level alignment. It attains the highest STAR in all four benchmark settings and with both Llama3.1-8B and Qwen3-8B. On the long-horizon DTELS-Bench, these gains also translate into the best Acc overall. For example, with Qwen3-8B at $K = 5$, Time-GRPO reaches 0.631 in STAR and 0.447 in Acc, compared with 0.507/0.388 for SFT and 0.572/0.407 for DPO. On CCKS2025, the most consistent gain remains STAR, while exact-match accuracy is more competitive. This pattern suggests that Time-GRPO is most useful when timeline transformation must preserve temporal-semantic evolution over long spans rather than optimize only local lexical overlap.

Online Policy Optimization vs. Same-Reward Imitation. The comparison with RS-SFT isolates the contribution of online policy optimization. RS-SFT uses the same composite reward to construct pseudo targets, but it does not optimize the policy online. Its gap to Time-GRPO therefore cannot be explained by reward design alone. In some burst-heavy settings, RS-SFT remains competitive on ROUGE. For instance, on CCKS2025 with Qwen3-8B at $K = 5$, RS-SFT achieves a higher R-1 than Time-GRPO (0.293 vs. 0.260). Even in this case, however, Time-GRPO is substantially better on STAR (0.668 vs. 0.448) and modestly better on Acc (0.173 vs. 0.147). These results indicate that online policy optimization is more effective than same-reward imitation at handling the discrete trade-offs among coverage, pacing, and redundancy.

Regime-Dependent Behavior. The relative strength of different methods depends on the temporal regime. On CCKS2025, where salient events are densely concentrated within a short horizon, HC-Centroid remains a strong baseline for exact-match node selection, reaching the highest Acc at both $K = 5$ and $K = 10$ (0.260 and 0.446). This suggests that explicit clustering is a strong inductive bias when local burst anchors dominate the task. On DTELS-Bench, however, Time-GRPO is clearly stronger. It surpasses HC-Centroid in both STAR and Acc, indicating that long-horizon TGT requires more than local density cues. In this regime, the main challenge is to preserve temporal-semantic evolution across alternating burst and dormant phases, which is precisely where sequence-level reasoning is most helpful.

4.3 Ablation Study. As summarized in Table 4.4, the removal of key components validates the design of **Time-GRPO**. The results yield the following insights:

- **Impact of R_{STAR}:** Removing the Semantic-Temporal Alignment Reward leads to the sharpest decline in narrative logic (STAR: $0.631 \rightarrow 0.548$), confirming its role as the linchpin in aligning semantic transitions with temporal spans.
- **Necessity of R_{Time} and R_{Info}:** Ablating either the density-aware temporal reward (R_{Time}) or entropy-regularized information reward (R_{Info}) results in significant degradation in semantic coverage, with R-1 scores dropping by 0.069 and 0.060, respectively. This underscores their importance in capturing the evolutionary dynamics of event streams.
- **Curriculum Learning & Stability:** Interestingly, the removal of Adaptive Curriculum Learning yields comparable final metrics in this static snapshot. However, these scalar values mask the underlying optimization volatility. Without the curriculum, the training process is prone to reward collapse and extreme variance. We visualize this phenomenon and discuss why curriculum learning is indispensable for robust convergence in the following **Training Stability Analysis**.

4.4 Training Stability Analysis. To validate stability, we present 5-seed statistics (Table 4.5) and visualize the trajectory of the best run (Figure 4.1).

Volatility in Training. Table 4.5 exposes the severe performance volatility of Time-GRPO without Curriculum Learning ($\sigma^2 = 0.022$, Min Acc=0), mirroring the erratic trajectory in Figure 4.1. Absent Curriculum Learning, the policy fails to reconcile hard structural constraints with soft narrative goals, resulting in optimization oscillations that disrupt stable gradient estimation.

Table 4.4: Ablation results on DTELS-Bench ($K = 5$, Qwen3-8B). w/o denotes the removal of a specific component.

Variant	Acc	R-1	STAR
Time-GRPO (Full)	0.447	0.373	0.631
w/o R_{STAR}	0.408	0.315	0.548
w/o R_{Time} (TPP)	0.414	0.304	0.628
w/o R_{Info} (IB)	0.419	0.313	0.609
w/o Curriculum Learning	0.445	0.375	0.633

Table 4.5: Stability Analysis over 5 Random Seeds. Despite similar average scores, the baseline suffers from extreme volatility and **complete training collapse** (Min Acc = 0), whereas Time-GRPO ensures robust convergence.

Method	Reward Var. (σ^2)	Acc (Avg$\pm$Std) (Robustness)	Min Acc (Worst Case)
Time-GRPO	**0.005**	**0.447 $\pm$ 0.015**	**0.435**
w/o Curriculum	0.022	0.445 $\pm$ 0.171	0

Stabilization via Curriculum. Time-GRPO achieves robust convergence ($\sigma^2 = 0.005$). As shown in Figure 4.1 (blue solid line), the Adaptive Curriculum Learning strategy orchestrates a distinct evolution:

- **Phase I (Format Alignment, Steps 0-400):** The reward curve initially dips and stabilizes, indicating the model is effectively constrained to the valid action subspace, pruning out high-variance “hallucinations.”
- **Phase II and III (Refinement, Steps 400+):** As weights for narrative logic (R_{STAR}) are activated, the reward exhibits a steady upward trend. The curriculum smooths the optimization landscape, guiding the policy from structural correctness to high-level semantic reasoning without the collapse observed in the no-curriculum ablation.

This contrast shows that adaptive curriculum learning is important for robust convergence, even though the mean final metrics are similar in this ablation snapshot. Due to space constraints, we defer the generalization analysis and the comprehensive case studies to Appendix D and Appendix F.

5 Related Work.

5.1 Timeline Summarization. Timeline summarization (TLS) aims to construct chronologically ordered summaries for long-running events. Early work relied on graph-based ranking and submodular optimization to balance salience and redundancy [1], while later approaches introduced neural architectures and event-centric modeling to better capture cross-document and temporal dependencies [13, 4, 22, 11]. Despite these advances, most TLS methods still generate a single timeline from a document collection and treat event selection mainly as a salience-and-

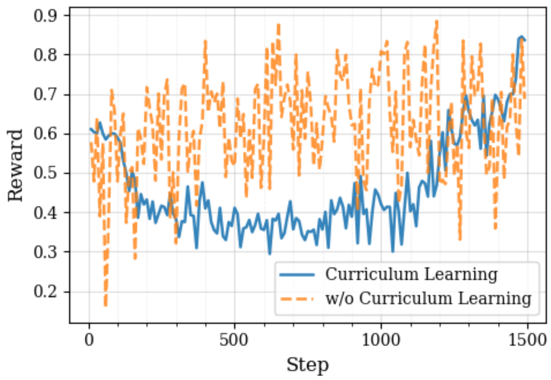

Figure 4.1: **Comparison of training reward trajectories.** The standard training (orange dashed line) exhibits extreme volatility due to the sparse rewards in the discrete search space. The proposed Adaptive Curriculum Learning (blue solid line) effectively mitigates optimization instability, leading to smoother convergence and superior asymptotic performance.

coverage problem. They are not designed for post-hoc transformation of an already available timeline under an explicit budget.

5.2 Dynamic Granularity and Controlled Timeline Summarization. The line of work most closely related to ours is dynamic-granularity timeline summarization. DTELS [32] shows that the same topic may admit multiple valid levels of detail and studies timeline generation from source documents conditioned on a granularity indicator. Our setting is different. TGT takes an existing fine-grained timeline as input and transforms it into one target K-node timeline at inference time. The output must satisfy exact-cardinality and strict-subsequence constraints, so the task is not free-form multi-granularity generation from raw documents.

TGT also differs from constrained timeline summarization [18]. In constrained TLS, the constraint specifies which aspect or perspective of a topic should be summarized from the document set. In TGT, the control signal specifies the global granularity of the full timeline. The goal is therefore to compress the whole event trajectory into a coarser but faithful representation, rather than summarize one thematic slice. More broadly, controllable text generation often regulates surface attributes such as length or density through control codes or decoding constraints [7, 12]. TGT instead requires sequence-level compression that preserves temporal-semantic evolution.

5.3 Temporal Modeling and Policy Optimization. Real event streams are non-homogeneous, often alternating between bursty periods and long dormant periods. Classical clustering methods and timeline models capture temporal structure to some extent [23, 15, 22], but they do not explicitly optimize the alignment between temporal spans and semantic change under a hard budget. Point-process models provide a useful view of bursty dynamics [5, 10, 21], and information-bottleneck ideas offer a principled view of compression [25, 8]. We use these lines differently: point-process intuition motivates our density-aware temporal reward, while the information objective is used to balance coverage and redundancy in timeline transformation rather than event prediction [14].

Because TGT has sequence-level goals but no canonical intermediate traces, reinforcement learning is a natural fit. Building on prior work on discrete-metric optimization and reward-based summarization [17, 24], we adopt Group Relative Policy Optimization (GRPO) [20] to optimize candidate timelines without a separate critic network, and combine it with adaptive curriculum learning [3] for stable training.

6 Conclusion. We study **Timeline Granularity Transformation (TGT)**, a budget-conditioned timeline-to-timeline transformation problem in which a fine-grained source timeline is compressed into a shorter target timeline under exact-cardinality and strict-subsequence constraints. The central challenge is to preserve temporal-semantic evolution under non-homogeneous event dynamics, where burst phases and dormant phases make local salience ranking or near-uniform spacing insufficient.

To address this problem, we propose **Time-GRPO**, a sequence-level reinforcement learning framework that combines three rewards. **STAR** encourages alignment between semantic change and temporal span. A **density-aware temporal reward** adapts selection to non-uniform event distributions. An **entropy-regularized information reward** balances coverage and redundancy under tight budgets. We further introduce an **adaptive curriculum** to stabilize optimization in the discrete combinatorial search space.

Experiments on DTELS-Bench and CCKS2025 at the benchmark-supported budgets $K = 5$ and $K = 10$ reveal a consistent pattern. Time-GRPO achieves the strongest STAR across datasets and backbones, and it attains the best Acc among the open-source LLMs we evaluate. Comparisons with RS-SFT show that these gains do not come from reward design alone; online policy optimization is more effective than same-reward imitation at handling the trade-offs among coverage, pacing, and redundancy. The gains are largest on long-horizon timelines, while burst-heavy streams remain a boundary case in which explicit clustering is still highly competitive on exact-match node selection. The adaptive curriculum also improves robustness by reducing optimization volatility across runs.

Overall, the results suggest that sequence-level reinforcement learning is a promising approach to budget-conditioned timeline transformation, especially for long-horizon, non-homogeneous event streams. Future work includes evaluation at unseen budgets with additional annotations or preference signals, multimodal event streams, and incremental or streaming timeline transformation.

A Mathematical Derivation of GRPO. In this section, we will present the mathematical formulation of the Group Relative Policy Optimization (GRPO) used in our **Time-GRPO** framework.

A.1 Group-Based Advantage Estimation. To estimate the advantage without a critic network, we sample a group of G candidate timelines $\{\mathcal{G}_K^{(1)}, \ldots, \mathcal{G}_K^{(G)}\}$ from the policy π_θ for each input $\mathcal{G}_N$. The advantage A_i corresponding to the i-th candidate is then computed by normalizing its composite reward against the group:

$$A_i = \frac{R(\mathcal{G}_K^{(i)}|\mathcal{G}_N) - \mu_R}{\sigma_R + \epsilon}, \tag{A.1}$$

where μ_R and σ_R are the mean and standard deviation of the intra group rewards $\{R_1, \ldots, R_G\}$, and ϵ is a small constant for numerical stability. This relative comparison reduces the high variance from discrete subset selection.

A.2 Optimization Objective. Time-GRPO governs the policy update trajectory by maximizing a composite surrogate objective, $\mathcal{L}_{GRPO}(\theta)$, which synthesizes importance sampling with trust-region constraints to mitigate optimization volatility:

(A.2)

$$\mathcal{L}_{GRPO}(\theta) = \mathbb{E}_{q\sim D}\left[\frac{1}{G}\sum_{i=1}^{G}\left(\min\Big(r_i(\theta)A_i, \mathrm{clip}\big(r_i(\theta), 1-\beta, 1+\beta\big)A_i\Big) - \lambda D_{KL}(\pi_\theta||\pi_{ref})\right)\right],$$

where $r_i(\theta) = \frac{\pi_\theta(\mathcal{G}_K^{(i)}|\mathcal{G}_N)}{\pi_{\theta_{old}}(\mathcal{G}_K^{(i)}|\mathcal{G}_N)}$ denotes the probability ratio between the current and old policies. The function $\mathrm{clip}(\cdot)$ restricts the policy update within the range $[1-\beta, 1+\beta]$, and λ controls the strength of the KL regularization to suppress excessive deviation from the reference model π_{ref}.

B Implementation Details and Hyperparameters. To ensure the reproducibility of our experiments, we provide the detailed mathematical formulations of our reward components, followed by the computational infrastructure, model configurations, and specific training hyperparameters used for **Time-GRPO**.

B.1 Details of TPP-Inspired Distribution Reward. In this subsection, we detail the mathematical formulation of the Multi-Indicator Voting Mechanism used to categorize timeline heterogeneity, and the differentiable implementation of the temporal reward functions.

B.1.1 Multi-Indicator Voting Mechanism. To robustly categorize the source timeline $\mathcal{G}_N$ into a *Uniform State* ($\mathcal{U}$) or a *Clustered State* ($\mathcal{C}$), we employ a voting mechanism based on three dispersion indices derived from the sorted timestamp intervals $\Delta T = \{\delta_1, \delta_2, \ldots, \delta_{N-1}\}$, where $\delta_i = t_{i+1} - t_i$.

Dispersion Metrics We utilize three metrics to capture distinct aspects of temporal distribution:

- **Coefficient of Variation (CV):** Measures relative dispersion.

$$CV = \frac{\sigma_{\Delta T}}{\mu_{\Delta T} + \epsilon}. \tag{B.1}$$

- **Max-Min Ratio (R):** Measures the extremity of temporal gaps to detect outliers.

$$R = \frac{\max(\Delta T) + \epsilon}{\min(\Delta T) + \epsilon}. \tag{B.2}$$

- **Zero-Interval Ratio (ZR):** Measures the intensity of simultaneous event bursts.

$$ZR = \frac{1}{N-1}\sum_{i=1}^{N-1}\mathbb{I}(\delta_i = 0), \tag{B.3}$$

where $\epsilon = 10^{-8}$ is a smoothing term for numerical stability.

Voting Logic We define a tiered scoring system to determine the uniformity level. Each metric contributes to a cumulative score S based on thresholds empirically calibrated on the validation set, as shown in Table B.1.

Table B.1: Thresholds for Multi-Indicator Voting Mechanism.

Metric	Strong (+2 pts)	Weak (+1 pt)	Non-Uniform (+0 pts)
CV	< 0.6	$[0.6, 1.2]$	> 1.2
R	< 8	$[8, 15]$	> 15
ZR	< 0.2	$[0.2, 0.4]$	> 0.4

The final state is determined by the total score $S \in [0, 6]$. We define the state of the source timeline $\mathcal{G}_N$ as:

$$\mathrm{State}(\mathcal{G}_N) = \begin{cases} \mathcal{U} & \text{if } S \geq 5, \\ \mathcal{C} & \text{otherwise.} \end{cases} \tag{B.4}$$

B.1.2 Differentiable Reward Implementation The temporal reward R_{Time} is computed as a weighted sum of an alignment component and an adaptive component:

$$R_{\text{Time}} = 0.6 \cdot R_{\text{align}} + 0.4 \cdot R_{\text{adapt}}. \tag{B.5}$$

Alignment Reward (R_{align}) This component ensures the generated timeline $\mathcal{G}_K$ follows the temporal pacing of the ground truth $\mathcal{G}_{\text{Ref}}$. It combines span consistency and interval similarity:

$$R_{\text{align}} = 0.5 \cdot \left(1 - \left|\frac{\text{Span}(\mathcal{G}_K)}{\text{Span}(\mathcal{G}_{\text{Ref}})} - 1\right|\right) + 0.5 \cdot \text{CosSim}(\Delta T_K, \Delta T_{\text{Ref}}), \tag{B.6}$$

where $\text{CosSim}(\cdot)$ denotes the cosine similarity between the interval vectors, normalized to $[0, 1]$.

Adaptive Distribution Reward (R_{adapt}) The objective of this component switches dynamically based on the detected state:

- **Case $\mathcal{U}$ (Uniform Source):** The model is rewarded for low variance.
(B.7)
$$R^{\mathcal{U}}_{\text{adapt}} = 0.5 \cdot \text{Cov} + 0.5 \cdot \left(1 - \min\left(1, \frac{CV(\mathcal{G}_K)}{0.6}\right)\right).$$
- **Case $\mathcal{C}$ (Clustered Source):** The model is rewarded for replicating the burstiness (Zero-Interval Ratio) of the source.
(B.8)
$$R^{\mathcal{C}}_{\text{adapt}} = 0.6 \cdot \text{Cov} + 0.4 \cdot (1 - |ZR(\mathcal{G}_K) - ZR(\mathcal{G}_N)|).$$

Here, $\text{Cov} = \min(1, \text{Span}(\mathcal{G}_K)/\text{Span}(\mathcal{G}_{\text{N}}))$ represents the global temporal coverage ratio.

B.2 Detailed Implementation of Information Reward. In this subsection, we provide the precise mathematical operationalization of the Entropy-Regularized Information Reward (R_{Info}) defined in Eq. (3.6). While the theoretical objective is formulated as a composite trade-off, our practical implementation employs a multiplicative gating mechanism and an adaptive entropy constraint.

Specifically, we decompose the operationalization of R_{Info} into three primary components: the repetition control mechanism, the adaptive entropy normalization, and the final composite integration.

Repetition Gating Factor. To strictly minimize redundancy within the constrained budget, we calculate a gating factor based on the count of duplicate event summaries. Let $\mathcal{S} = \{s_1, s_2, \ldots, s_K\}$ be the list of selected event summaries in the distilled timeline $\mathcal{G}_K$. We first define the repetition count N_{rep} as:

$$N_{\text{rep}} = \sum_{u \in \text{unique}(\mathcal{S})} (\text{count}(u) - 1), \tag{B.9}$$

where $\text{count}(u)$ denotes the frequency of summary u.

The repetition gating factor $\mathcal{M}_{\text{Rep}} \in [0, 1]$ is derived as a linear dampening coefficient:

$$\mathcal{M}_{\text{Rep}} = \max\left(0, 1 - \frac{N_{\text{rep}}}{K}\right). \tag{B.10}$$

If the timeline contains no duplicates ($N_{\text{rep}} = 0$), $\mathcal{M}_{\text{Rep}} = 1.0$. Conversely, as redundancy increases, the factor decays linearly, strictly penalizing the effective information gain.

Adaptive Normalized Entropy. To ensure the lexical diversity score is robust to varying timeline complexities (e.g., simple incidents vs. complex geopolitical events), we adopt an **adaptive normalization strategy**.

Let $H(\mathcal{G}_K)$ be the bigram entropy of the generated timeline, and $H(\mathcal{G}_{\text{Ref}})$ be the bigram entropy calculated from the corresponding gold-standard reference timeline. The normalized entropy score is defined as the ratio of the generated entropy to the reference entropy, capped at unity:

$$H_{\text{norm}} = \min\left(1, \frac{H(\mathcal{G}_K)}{H(\mathcal{G}_{\text{Ref}}) + \epsilon}\right), \tag{B.11}$$

where $\epsilon = 10^{-8}$ ensures numerical stability. This formulation creates a dynamic baseline: if the distilled timeline achieves a lexical richness level comparable to or exceeding that of the human-annotated reference, H_{norm} saturates at 1.0. Otherwise, it quantifies the relative preservation of information density.

Composite Score Calculation. The final R_{Info} is computed as a weighted sum of the *gated relevance* and *adaptive diversity*. We apply word segmentation (via Jieba) before calculating metrics to handle linguistic features.

The operational formula is:
(B.12)
$$R_{\text{Info}} = 0.8 \cdot (\text{ROUGE-1}(\mathcal{G}_K, \mathcal{G}_N) \times \mathcal{M}_{\text{Rep}}) + 0.2 \cdot H_{\text{norm}}.$$

Comparing this instantiation to the general form in Eq. (3.6), we set the balancing coefficients as $\mu_1 = 0.8$ and $\mu_2 = 0.2$. This configuration prioritizes semantic coverage grounded by non-redundancy, while reserving 20% of the reward budget to encourage lexical diversity relative to the ground truth.

B.3 Experimental Environment and Model Configuration. All experiments were conducted on a computational node equipped with $4 \times$ NVIDIA A800 (48GB) GPUs. The implementation is based on the PyTorch framework and the Hugging Face *Transformers* library. To optimize memory efficiency during the reinforcement learning phase, we employed bfloat16 precision and enabled gradient checkpointing.

We utilized **Llama-3.1-8B-Instruct** and **Qwen-3-8B-Instruct** as the base policy networks. And we applied Low-Rank Adaptation (LoRA) to all linear layers to navigate the optimization landscape while constraining the trainable parameters. Table B.2 presents the specific LoRA configuration.

B.4 Training Hyperparameters. The specific hyperparameters for the Time-GRPO training phase are listed in Table B.3. We observed that Qwen3-8B required a slightly

Table B.2: LoRA Configuration for both models.

Parameter	Value
Rank (r)	8
Alpha (α)	16
Dropout	0.05
Bias	None
Target Modules	`[q, k, v, o, gate, up, down]_proj`
Task Type	CAUSAL_LM

lower warmup ratio for optimal convergence compared to Llama-3.1-8B.

Table B.3: Training hyperparameters for Time-GRPO. The Group Size ($G = 8$) indicates that 8 candidate timelines are sampled for each prompt during the group relative advantage estimation.

Hyperparameter	Llama-3.1-8B	Qwen-3-8B
Precision	bfloat16	bfloat16
Max Sequence Length	8192	8192
Max Completion Length	1024	1024
Per-Device Batch Size	1	1
Group Size (G)	8	8
Gradient Accumulation	8	8
Optimization	AdamW	AdamW
Learning Rate	1e-5	2e-5
LR Scheduler	Cosine	Cosine
Warmup Ratio	0.05	0.03
Num Epochs	5	5

B.5 Adaptive Curriculum Learning Schedule. We implement the curriculum learning strategy by dynamically modulating the reward coefficients $\omega_c(t)$ via a smooth Sigmoid transition function. This formulation ensures stable gradient updates as the optimization objective evolves from low-level constraints to high-level narrative logic:

$$\omega_c(t) = \omega_{\text{start}} + \frac{\omega_{\text{end}} - \omega_{\text{start}}}{1 + e^{-k\cdot(t-\tau_c)}}, \tag{B.13}$$

where T_{max} denotes the total training steps, and $k = 0.02$ governs the smoothness of the transition. We define two critical phase boundaries at $\tau_1 = 0.1T_{max}$ and $\tau_2 = 0.6T_{max}$, corresponding to the completion of structural adaptation and semantic anchoring, respectively.

The specific trajectories for each reward component are configured as follows:

- **Structural Stabilization ($R_{\textbf{Format}}$):** To prune the invalid action space early in training, ω_{Format} initializes at a dominant weight of 0.5. Post-τ_1, it anneals to a maintenance level of 0.05 to prevent policy collapse while allowing exploration for semantic rewards ($0.5 \to 0.05$).
- **Informativeness Anchoring ($R_{\textbf{Info}}$):** The weight for semantic content increases monotonically from 0.2 to 0.6 centered at τ_1. To prioritize factual correctness over stylistic imitation, we decompose the total informativeness weight Ω_{Info} into date accuracy and textual fidelity:

$$\omega_{\text{Date}} = 0.7 \cdot \Omega_{\text{Info}}, \quad \omega_{\text{Text}} = 0.3 \cdot \Omega_{\text{Info}}. \tag{B.14}$$

  This results in effective ranges of $[0.14, 0.42]$ and $[0.06, 0.18]$, respectively.
- **Narrative Refinement ($R_{\textbf{STAR}}, R_{\textbf{Time}}$):** High-level reasoning rewards are introduced during the final refinement phase (τ_2), transitioning from a negligible 0.05 to 0.35. The logic budget is distributed equally to synchronize temporal pacing with density modeling:

$$\omega_{\text{STAR}} = \omega_{\text{Time}} = 0.5 \cdot \Omega_{\text{Logic}}, \tag{B.15}$$

  yielding a final weight of 0.175 for each component.

B.6 Hyperparameter Sensitivity and Justification. A potential concern in reinforcement learning frameworks involving composite rewards is the sensitivity to hyperparameter settings, particularly the weighting coefficients in the curriculum learning schedule. We tackle this issue by clarifying the rationale behind our choices and demonstrate the robustness of **Time-GRPO**.

Empirical Stability. The hyperparameters presented in Table B.3 and the curriculum schedule were determined based on a coarse-to-fine grid search on a validation set. Crucially, we observe that Time-GRPO is **not sensitive to precise values** but rather to the *relative order* of phase transitions. For instance, varying the phase transition boundaries (τ_1, τ_2) by $\pm 10\%$ or adjusting the final weight ratios (e.g., $\omega_{Date}/\omega_{Text}$) within a reasonable range ($0.6/0.4$ to $0.8/0.2$) results in marginal performance fluctuations (less than 1.5% in STAR and ROUGE scores). This indicates that the efficacy of our method stems from the *curriculum learning paradigm itself*—which progressively shifts focus from syntactic correctness to semantic logic—rather than specific, over-tuned scalar values.

Heuristic Logic. The dynamic weights follow a logical progression rather than arbitrary tuning:

- **Phase I (Format):** A high initial penalty is strictly necessary to prune invalid actions in the large search space $\binom{N}{K}$.
- **Phase II (Anchoring):** Once structural validity is met, the model must anchor to ground-truth milestones (High R_{Info}) before attempting complex reasoning.
- **Phase III (Reasoning):** Narrative logic (R_{STAR}) is a high-level property that can only be optimized once the model selects semantically relevant nodes.

Future Directions. While the current static scheduling proves effective and robust, we acknowledge that manual tuning of $\omega(t)$ can be labor-intensive. Future work will explore **Auto-Curriculum Learning**, employing meta-learning or multi-objective gradient descent (e.g., MGDA) to dynamically adjust reward weights based on the variance of the group advantage, thereby eliminating the need for manual hyperparameter specification.

C Relationship Between STAR and Acc This appendix examines the empirical relationship between the Semantic-Temporal Alignment Reward (STAR) and Node Accuracy (Acc). Our goal is not to claim that STAR fully captures overall timeline quality. Instead, we test whether higher STAR is associated with better task performance and whether STAR provides information beyond exact-match accuracy.

C.1 Experimental Setup We aggregated generation outputs from DTELS-Bench at $K = 5$ across a diverse set of models, including the proprietary GPT-5.2 and open-source models under both SFT and Time-GRPO settings. Using the results from our 5-fold cross-validation, we collected a total of $N = 1{,}865$ instance-level sample pairs $(STAR_i, Acc_i)$. This sampling strategy reduces the chance that the observed pattern is an artifact of a single model or optimization setting.

C.2 Statistical Relationship and Interpretation To quantify linear and monotonic dependencies, we compute the Pearson (r) and Spearman rank (ρ) coefficients:

- **Pearson correlation:** $r = 0.20$ ($p < 0.001$)
- **Spearman correlation:** $\rho = 0.11$ ($p < 0.001$)

These results support two observations.

Positive but modest association. The positive coefficients and low p-values indicate that STAR is not independent of task performance: higher STAR is, on average, associated with higher Acc. This reduces the concern that optimizing STAR is entirely disconnected from the task objective.

Complementary rather than equivalent. The effect size is modest, especially at the instance level. This suggests that STAR should not be interpreted as a standalone proxy for overall quality. Instead, it captures a complementary sequence-level property. Acc measures point-wise correctness, i.e., whether the selected nodes exactly match the reference. STAR measures whether semantic change is paced in a way that is consistent with temporal span. The two metrics therefore evaluate related but different aspects of timeline quality.

Taken together, these statistics suggest that STAR is a useful complementary metric for temporal-semantic pacing, rather than a complete substitute for exact-match evaluation or human judgment.

C.3 Trend Analysis Because instance-level correlations are naturally noisy across diverse models and outputs, we also perform a binning analysis to visualize the aggregate relationship between STAR and Acc. We again use the $N = 1{,}865$ samples from DTELS-Bench at $K = 5$.

Figure C.1 groups the samples into five STAR quantiles. The grouped trend is positive overall: average accuracy increases from **0.384** in the "Low" bin to **0.514** in the "High-Mid" bin. This pattern suggests that higher STAR is generally associated with better Acc at the aggregate level.

At the same time, the highest STAR bin shows a slight drop (**0.472**). This suggests that maximizing temporal-semantic alignment alone does not guarantee the best exact-match accuracy on every instance. The result is consistent with our use of a composite reward: STAR encourages sequence-level pacing, while R_{Info} and R_{Time} help balance coverage, anchoring, and redundancy.

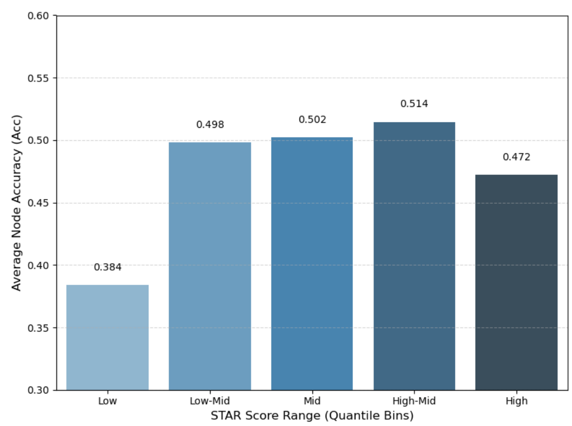

Figure C.1: **Average Acc across STAR score intervals.** Aggregated results over $N = 1{,}865$ samples from DTELS-Bench at $K = 5$. While instance-level correlation is modest ($r = 0.20$), the grouped analysis shows a positive overall trend: bins with higher STAR tend to have higher average accuracy. The slight dip in the highest bin suggests that STAR is best interpreted as a complementary sequence-level metric rather than a standalone objective.

D Methodological Versatility: Narrative Turning Point Identification. To demonstrate that **Time-GRPO** is a general paradigm rather than a dataset-specific solution, we evaluate its extensibility on **Turning Point (TP) Identification** within the TRIPOD dataset [16]. This task requires the model to transcend explicit timestamps and capture the latent narrative logic of stories. This transition from news to narratives validates the robustness of our framework across diverse sequence modeling tasks.

D.1 Dataset and Task Formulation. We utilize the TRIPOD dataset [16], which contains movie synopses (averaging 35 sentences) and requires the identification of exactly $K = 5$ thematic TPs: *Opportunity*, *Change of Plans*, *Point of No Return*, *Major Setback*, and *Climax*. We formulate this as a Timeline Granularity Transformation task, where the goal is to extract the most representative $\mathcal{G}_K$ from the full synopsis $\mathcal{G}_N$.

D.2 Logical Time Mapping via Cumulative Text Length. To bridge the gap between narrative progression and our temporal-aware reward functions, we define a **Logical Time** t_i for each sentence c_i. We adopt a *Cumulative Text Length Mapping* strategy, assuming that narrative progress is proportional to the amount of information conveyed. For a synopsis with N sentences, the logical time t_i of the i-th sentence is defined as:

$$t_i = \frac{\sum_{j=1}^{i} \text{len}(c_j)}{\sum_{j=1}^{N} \text{len}(c_j)}, \tag{D.1}$$

where $\text{len}(c_j)$ denotes the character count of sentence j. This mapping normalizes the plot into a logical event stream $t_i \in [0, 1]$, enabling the direct application of **Time-GRPO**'s temporal constraints.

Table D.1: Generalization results on TRIPOD (Movie Synopses). TAM is the domain-specific SOTA. **N/A** indicates metrics not applicable/reported as TAM predicts positions rather than generating text.

Method	Acc	R-1	R-2	STAR
Time-GRPO	0.453	0.664	0.523	0.768
GPT-5.2	0.440	0.628	0.479	0.733
HC-Centroid	0.214	0.468	0.236	0.735
TAM + entities	0.413	-	-	-

D.3 Experimental Results. We benchmark Time-GRPO against a methodological spectrum spanning statistical clustering (HC-Centroid) and zero-shot reasoning (GPT-5.2). Table D.1 shows **Time-GRPO** outperforms prior SOTA in turning point accuracy and narrative consistency. A closer analysis of these results reveals two key insights:

- **Rhythm Perception**: The **STAR** reward successfully guides the model to align significant “Semantic Shifts” with appropriate “Logical Time” spans, preventing the model from picking TPs that are semantically redundant or temporally clustered.
- **Structural Adaptation**: The TPP-based distribution prior effectively regularizes the model to follow the “Three-Act Structure” inherent in dramatic narratives, proving that the temporal heuristics learned in the news domain are transferable to literary logic.

These findings confirm that **Time-GRPO** is not merely a news summarization tool but a universal framework for constrained narrative reasoning and granularity transformation.

E Prompts. To facilitate reproducibility, we provide the exact instruction templates used for the Policy Network (π_θ) across different tasks. We employ a consistent prompt structure that defines the role, task objectives, strict constraints, and output format.

E.1 Prompt for Timeline Granularity Transformation (News). Table E.1 outlines the system instruction for the main task, which involves converting fine-grained news event timelines to a different granularity.

E.2 Prompt for Narrative Turning Point Identification (Movies). To test the generalization capability of Time-GRPO, we adapted the prompt for the TRIPOD dataset (movie synopses). As shown in Table E.2, this task requires strict adherence to narrative structure definitions.

Table E.1: System Instruction for TGT.

System Instruction:
You are a professional news editor assistant. Your task is to select the most important **K** key event summaries from the given news topic timeline ($\mathcal{G}_N$) to help readers quickly grasp the core development and evolutionary pulse of the event.
Rules:
1. Select events ONLY from the provided `timeline` list. **Do NOT fabricate or modify the original summary content.**
2. Prioritize events that are **iconic, turning points, widely concerned by the public, or cause significant subsequent impact**.
3. Output in `<answer>JSON</answer>` format. The JSON must contain `title` and `top_k_timeline` fields, where `top_k_timeline` is a list containing K objects. Each object must include `time` and `summary` fields.
4. Maintain chronological order (ascending by `time`).
Output Format Example:
$< answer >$ { "title": "News Title", "top_k_timeline": [{"time": "Time 1", "summary": "Summary 1"}, {"time": "Time 2", "summary": "Summary 2"}, ... (Total K items)] } $< /answer >$
Input Timeline:
Timeline

F Case Study. To qualitatively validate the effectiveness of Time-GRPO in handling temporal non-homogeneity, we present a detailed case study on the topic “*The Disappearance of MH370*” from the DTELS-Bench dataset. This trajectory is highly irregular, manifesting as a bimodal burst pattern: a prolonged **Dormant Phase** is sandwiched between an initial **Burst Phase** (2014) and a subsequent hyper-dense **Burst Phase** (2023).

Table E.2: System Instruction for Movie Turning Point Identification. The model is tasked with identifying 5 specific structural milestones in a narrative arc.

System Instruction:
Your task is to EXTRACT exactly **K** key **Turning Points (TPs)** from the provided movie synopsis timeline. You must act as a precise data extractor to help readers grasp the core dramatic arc of the story by selecting the most critical events.
Strict Rules:
1. **Definition Alignment**: You must identify and select one event for each of the following 5 stages: - **TP1 (Opportunity)**: The introductory event where the protagonist is first pulled into the plot. - **TP2 (Change of Plans)**: The pivot point where the story's main goal is defined. - **TP3 (Point of No Return)**: The event where the character fully commits to the goal. - **TP4 (Major Setback)**: The critical low point where the initial plan fails. - **TP5 (Climax)**: The ultimate confrontation and resolution of the narrative arc.
2. **Strict Extraction (Critical for Evaluation)**: - Select events ONLY from the provided `timeline`. - **Copy the `time` and `summary` fields EXACTLY as they appear in the source.** - Do NOT change any punctuation, do NOT rephrase, and do NOT summarize the summary. Any character difference will result in evaluation failure.
3. **Narrative Rhythm**: Ensure the selected nodes represent the full arc of the story from beginning to end, maintaining a balanced narrative flow.
4. **Output Format**: Response must be in <answer>JSON</answer> format. The JSON must include `title` and `top_k_timeline` fields.
5. **Chronological Order**: Maintain strictly ascending order based on the `time` field.
Output Format Example:
$< answer >$ { "title": "Movie Title", "top_k_timeline": [{"time": "Original Time 1", "summary": "Original Sentence for TP1"}, ... {"time": "Original Time 5", "summary": "Original Sentence for TP5"}] } $< /answer >$
Input Movie Synopsis:
Synopsis_Timeline

We compare the distillation results ($K = 5$) of **Time-GRPO** against a strong Baseline (Standard LLM with Uniformity Bias), as detailed in Table F.1.

F.1 Mitigating Narrative Erosion in Bursty Phases. The source timeline contains a critical burst of causal events immediately following the incident in 2014.

- **Baseline Failure (Uniformity Bias):** Driven by a rigid spacing heuristic, the Baseline distributes nodes evenly across the decade. Notably, as shown in Table F.1, it leaves a massive **3-year gap** between the initial contact loss (2014-03-08) and the search suspension (2017-01-17). This results in *Narrative Erosion*, completely omitting the dense search and rescue operations that are essential for understanding the context.
- **Time-GRPO Success (Density Awareness):** In contrast, Time-GRPO correctly identifies the initial burst density. It allocates two valuable node slots to the early phase (2014-03-08 and 2014-04-23), capturing the immediate causal aftermath despite the short temporal spans.

Table F.1: Comparison of Distilled Timelines ($K = 5$). The Baseline misses the dense initial phase. **Time-GRPO** captures the critical turning point in 2014 (bold) while ignoring redundancy.

Baseline (Uniformity Bias)		Time-GRPO (Ours)	
Date	**Event Summary**	**Date**	**Event Summary**
2014-03-08	MH370 flight lost contact.	2014-03-08	MH370 flight took off from KL Airport.
2017-01-17	China, Malaysia, Australia suspend search.	**2014-04-23**	**Search suspended due to bad weather.**
2018-07-30	Malaysia released 822-page report.	2016-02-19	Australia to end search; implies intentional crash.
2021-12-02	UK expert claims discovery of MH370.	2018-07-30	Malaysia released comprehensive report.
2023-11-27	Families' compensation case opened.	2023-11-16	Case scheduled to open session on Nov 27.

F.2 Robustness to Redundancy Noise. A critical challenge is the "recency bias" inherent in the source data, where the year 2023 contains an extreme cluster of 21 redundant events (Table F.2). This noise distribution exposes a sharp contrast in filtering capabilities between models:

- **The Trap:** Standard baselines statistically over-sample from this high-frequency cluster to maximize temporal coverage, often wasting the budget on trivial reaction events.
- **Time-GRPO's Resilience:** Despite 2023 accounting for exceeding 50% of source nodes, Time-GRPO selects only *one* representative node ("Case formally opens"). It effectively filters out 20 redundant entries, reserving the limited budget (K) to recover the root causes in 2014.

F.3 Semantic & Causal Consistency. Table F.1 provides a side-by-side comparison of the generated content. The **Baseline** creates a disjointed narrative by jumping from the initial loss of contact (2014-03-08) directly to the 2017 search suspension, omitting the dense early search-and-rescue developments in 2014. By contrast, Time-GRPO preserves the early event progression by allocating two nodes to the 2014 burst phase, including the weather-related suspension during the early search phase on 2014-04-23, thereby maintaining a more coherent temporal-causal narrative under extreme compression.

Mechanism Verification. The results confirm the efficacy of our composite rewards:

- **Density Adaptation:** The TPP-inspired reward successfully drove the high-density sampling in 2014, countering the uniform prior.
- **Pacing Alignment:** The STAR reward justified the large temporal leap from 2018 to 2023, as the semantic progression during the investigation hiatus was minimal.

F.4 Full Fine-grained Source Data. Table F.2 lists the complete fine-grained event timeline ($\mathcal{G}_N$) used in this case study. We can observe the **Real-world Noise** inherent in the data:

- **Redundancy:** Multiple updates occur on the same day (e.g., March 8, 2014, and November 17, 2023).
- **Irrelevance:** The timeline includes tangential events, for instance, the 2023 comedian's controversial remarks and speculative news about barnacles. For this task, the model needs to identify and ignore such irrelevant information.

G LLM Usage Disclosure. In this section, we disclose the use of Large Language Models (LLMs) in the preparation and polishing of this manuscript. We employed Google's Gemini series of tools exclusively for enhancing the writing quality of this manuscript. Specifically, LLMs were utilized to polish grammar, maintain writing consistency, and provide wording suggestions for key sections. These sections include the abstract, introduction, and related work. LLMs also assisted in brainstorming concise summaries of ablation results and conclusions. During the experimental phase, LLMs were occasionally consulted for suggestions on code adjustments and debugging.

All core contributions of the text, including the framework design, theoretical analysis, algorithm development, and experiments of **Time-GRPO**, were independently conceived and implemented by the authors. Content generated by LLMs has been carefully reviewed, and the authors take full responsibility for the content of this manuscript. All scientific innovations, argumentation, and experimental evaluations were completed independently by the authors.

Table F.2: The Origin Fine-grained Timeline ($\mathcal{G}_N$) for "The Disappearance of MH370".

Date	Event Summary	Date	Event Summary
2014-03-08	MH370 flight took off from Kuala Lumpur International Airport.	2023-11-16	MH370 case scheduled to formally open court session on Nov 27.
2014-03-08	MH370 flight lost contact.	2023-11-17	Lawyer interpreted the difficulty of rights protection.
2014-03-08	Malaysia Subang Air Traffic Control Centre confirmed the aircraft missing.	2023-11-17	Families responded to court opening: hoped for establishment of a search fund.
2014-03-08	China's Ministry of Transport initiated Level I emergency response.	2023-11-17	Many families of missing passengers received formal notifications.
2014-03-08	Seven countries dispatched rescue teams to the search area.	2023-11-17	Experts analyzed why the MH370 lawsuit took years to open.
2014-04-23	Search suspended due to bad weather in the search area.	2023-11-18	Plaintiff lawyer explained details: many difficulties still exist today.
2014-07-23	Australia restarted the search operation.	2023-11-20	The core demand of families remains finding the missing people.
2016-02-19	Australia to end MH370 search; claims disappearance might be intentional crash.	2023-11-21	Family member Jiang Hui: longing to dream of mother, hair turned gray.
2017-01-17	China, Malaysia, and Australia announced suspension of deep-sea search.	2023-11-21	Families: Malaysia closed the family communication center after the incident.
2018-01-10	Malaysia formally approved restarting the MH370 search.	2023-11-21	10 years for families: learning English, computers, and smartphones to find children.
2018-05-29	Search for MH370 Concludes; Malaysian Government Pledges to Release Investigation Report.	2023-11-21	Family member: "I am still waiting for my child to come home."
2018-07-30	Malaysia released the investigation report, totaling 822 pages.	2023-11-22	MH370 case opening soon; families put forward four major demands.
2018-11-30	The original MH370 investigation team was disbanded.	2023-11-27	Photographer's account: The vanished airline, the long sorrow.
2021-12-02	UK expert claimed discovery of MH370 under the sea.	2023-11-27	Families: Old Malaysia Airlines bankrupt, new one ignores us, no compensation.
2021-12-02	Missing passengers' relatives association: called for review of new analysis.	2023-11-27	MH370 passenger families' compensation case opened today.
2021-12-03	Families clarified rumors regarding "MH370 found".	2023-11-27	Families spoke out: The report hides many things.
2022-12-13	UK media revealed new investigation theory: potential criminal intent behind mystery.	2023-11-27	Over 110 families have signed settlement agreements.
2023-06-08	Stand-up comedian joked about MH370; Malaysian Foreign Minister criticized it.	2023-11-27	Case opens; some families still believe their loved ones are alive.
2023-07-05	The comedian became a columnist for Newsweek, claiming gratitude to Malaysia.	2023-11-28	Family member posted video response regarding current court progress.
2023-08-30	Barnacles found may help identify the location of MH370 crash.		

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